Neeraj Karoshi

Geospatial AI Engineer | Computer Vision | Remote Sensing & GIS | Foundation Models
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Enschede, Netherlands | Dehradun, India

Summary

Currently pursuing Master of Science - MS, GIS & EO with around 2.5 years of expertise in large-scale data processing, EDA, machine learning, and computer vision. Skilled in developing end-to-end AI pipelines including semantic segmentation, LULC change detection, and canopy height estimation using advanced tools such as PyTorch, GDAL, and LangGraph. Proven ability in scalable ETL automation, MLOps, and Deep Learning frameworks (Transformers, ViTs, LLMs), foundation model applications driving impactful satellite imagery analytics. Seeking to leverage core GeoAI skills to advance innovative remote sensing solutions in a dynamic environment.

Experience

Data Analyst 11/2023 – 07/2025
Transoft Solutions *Hyderabad, Telangana, India*

- Architected scalable ETL and analytics pipelines using Python, Pandas, SQL, and Scikit-Learn for large-scale enterprise datasets
- Developed supervised machine learning models for predictive analytics, operational optimization, and automated decision support
- Collaborated with engineering and business teams to optimize database operations and analytics infrastructure

Data Analyst Intern 08/2023 – 10/2023
United Nations Development Programme (UNDP) *New York, NY, USA*

- Automated economic intelligence workflows through API integrations, web scraping, and data ingestion pipelines
- Performed exploratory data analysis (EDA), statistical modeling, and regression analysis on multinational development datasets
- Built reproducible analytics workflows for policy-oriented reporting and visualization

Graduate Assistant 08/2022 – 12/2022
University of South Dakota *Vermillion, SD, USA*

- Handled research and teaching assignments, guiding undergraduates in various projects and machine learning concepts.
- Explored breakthroughs in Kernel Change Point Detection using advanced Deep Learning architectures (CNNs, RNNs) and GPU acceleration using CUDA for efficient time series segmentation and anomaly detection.

Software Developer 09/2021 – 01/2022
IIFL Finance *Mumbai, Maharashtra, India*

- Streamlined ETL and CRM automation workflows improving operational reporting efficiency
- Developed internal dashboards and automation utilities supporting business intelligence operations

Education

MSc Geo-information Science & Earth Observation (Geoinformatics) 09/2025 – Present
Indian Institute of Remote Sensing (IIRS), ISRO *Dehradun, India*
ITC Faculty Geo-Information Science and Earth Observation, University of Twente *Enschede, Netherlands*

Post Graduate Diploma in Computer Science & AI 2023 – 2024
IIIT Delhi *Delhi, India*

Bachelor of Engineering in Computer Science & Engineering 2017 – 2021
Sir M. Visvesvaraya Institute of Technology — Visvesvaraya Technological University *Bengaluru, Karnataka, India*

Projects

Road Extraction from Medium-Resolution LISS-IV Satellite Imagery

- Investigated the feasibility of automated road extraction from 5.8 m LISS-IV satellite imagery by developing deep learning segmentation pipelines for challenging medium-resolution remote sensing data.

LULC Classification using CLAY Foundation Model

- Fine-tuned CLAY foundation models for land-use/land-cover classification using Sentinel-2 imagery and transfer learning for Earth observation applications.

Canopy Height Estimation via DINOv3

- Developed a canopy height estimation pipeline leveraging the DINOv3 foundation model for feature extraction from high-resolution satellite imagery.

Flood Mapping using Sentinel-1 SAR

- Built an end-to-end SAR-based flood mapping pipeline using Sentinel-1 imagery, integrating speckle filtering, change detection, and flood inundation analysis.

GeoAI-Kit: Geospatial AI Toolkit

- Developing a geospatial AI toolkit to support reusable workflows combining geospatial data processing, remote sensing, and AI-based analysis. To be used as skill in claude code/codex.

Achievements

Winner – Space & Water Nexus Hackathon | *SATUrN Tech* October 2026

- Contributed to BruinRadar, a geospatial decision-support prototype combining satellite remote sensing, field boundaries, elevation, and rainfall data to identify potentially flooded potato fields and assess brown rot risk.

Technical Skills

Programming: Python, C/C++, SQL, Shell Scripting
AI & Deep Learning: PyTorch, TensorFlow/Keras, Scikit-learn, Computer Vision, Transfer Learning, Semantic Segmentation, Change Detection, Vision Transformers, Foundation Models (CLAY, DINOv3)
Geospatial & Remote Sensing: GDAL, Rasterio, GeoPandas, Xarray, rioxarray, PostGIS, QGIS, ArcGIS, Google Earth Engine, Sentinel-1 SAR, Sentinel-2, Optical/SAR Image Processing
EO Data Engineering: STAC APIs, Microsoft Planetary Computer, stackstac, odc-stac, Dask, Geospatial Data Cubes, Raster/Vector Processing, Automated Data Pipelines
AI Workflow Engineering: LangChain, LangGraph, LLM Tool Orchestration
Development Tools: Git, GitHub, Docker, Linux, Streamlit, REST APIs, Web Scraping